

Junior Backend Developer Interview Sheet

Candidate Name:

Date:

Stack: Python, MariaDB, JIRA, Confluence

Scoring:

0 = No understanding / wrong

1 = Basic understanding / minor errors

2 = Good understanding / can implement

3 = Excellent, clear explanation and practical knowledge.

Part 1: Python / Backend Basics

Question	Notes	Score
Explain WSGI. What is it and why is it important for Python web apps?	Look for understanding of request/response cycle.	
Difference between list, tuple, dictionary. When to use each?	Focus on immutability and key/value mapping.	
How do you handle exceptions in Python?	Candidate should demonstrate basic error handling.	
How would you normalize database data in Python? (Convert dates, decimals, etc.)	Should show conversion to string/float as needed.	
How do you hash a password before storing in DB? Why is this important?	Look for mention of bcrypt or similar.	

Part 2: Database / SQL

Question	Notes	Score
Write a SQL query to select all rows between two dates in "VehicleSalesTracker" table	Should use WHERE date_of_sale BETWEEN ... AND	
Explain stored procedures. Why use them instead of inline SQL in Python?	Should understand abstraction and security.	

What is COALESCE in SQL? How is it useful in update statements?	Used to update only if new value is provided.	
Prevent SQL injection when calling DB from Python.	Look for parameterized queries or stored procedures.	
Table has 50 columns. User wants to update 3 columns only. How do you handle it efficiently?	Expect dynamic update / COALESCE / stored procedure approach.	

Part 3: API / Backend Design

Question	Notes	Score
Explain GET, POST, PUT, DELETE. When to use each?	Candidate should explain semantics of HTTP methods.	
Difference between query parameters and JSON body in API requests?	JSON body = POST/PUT payload, query = URL params.	
Design a date filter API for VehicleSalesTracker (inputs, outputs).	Should mention date_from, date_to inputs and JSON output.	
How do you return errors from an API in JSON?	Example: { "status": "error", "message": "PE ref not found" }.	
Status codes 200, 400, 500. Give examples for each.	200 = success, 400 = invalid input, 500 = server error.	

Part 4: Tools / Soft Skills

Question	Notes	Score
How do you use JIRA to track tasks or bugs?	Look for creating stories, updating status.	
How do you use Confluence for documentation?	Should mention documenting APIs, workflows, decisions.	
Experience with DBeaver? How do you test queries safely?	Look for using test DB, SELECT before UPDATE.	
How do you debug backend issues?	Look for logging, exception handling, step-by-step.	
How do you organize code in a backend project?	Look for modules, functions, readability.	

Pass / Fail Recommendations:

Notes / Comments: